

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

hospital

local

online_shopping

customer_spending_report

high_value_products_report

mobile_report

orders

Welcome

_ mongosh...

_ mongosh...

customer_...

online_...

high_value...

orders

orders

>_MONGOSH

> use online_shopping

< switched to db online_shopping

> db.createCollection("orders")

< { ok: 1 }

> db.orders.insertMany([

{customer_id: "C101",

name: "Asha",

items: [

{ product: "Mobile", price: 15000 },

{ product: "Earphones", price: 2000 }

]

},

{

customer_id: "C102",

name: "Kiran",

items: [

{ product: "Mobile", price: 15000 },

{ product: "Laptop", price: 50000 }

]

},

{

customer_id: "C103",

name: "Neha",

```
> db.orders.aggregate([
  {$unwind: "$items"},
  {$match: {product: "Mobile",price: { $gte: 15000 }}}},
  {$group: {_id: "$product",totalCustomers: { $sum: 1 },
    totalRevenue: { $sum: "$price" },
    averagePrice: { $avg: "$price" },
    minimumPrice: { $min: "$price" },
    maximumPrice: { $max: "$price" },
    customerNames: { $push: "$name" },
    firstCustomer: { $first: "$name" }}}},
  {$project: {_id: 0,
    product: "Mobile",
    totalCustomers: 1,
    totalRevenue: 1,
    averagePrice: 1,
    minimumPrice: 1,
    maximumPrice: 1,
    customerNames: 1,
    firstCustomer: 1}},
  {$out: "mobile_report"}
])
```

```
<
> db.mobile_report.find()
< {
  _id: null,
  totalCustomers: 2,
  totalRevenue: 30000,
  averagePrice: 15000,
  minimumPrice: 15000,
  maximumPrice: 15000,
  customerNames: [
    'Asha',
    'Kiran'
  ],
  firstCustomer: 'Asha'
}
```

```
> db.orders.aggregate([
  {
    $unwind: "$items"
  },
  {
    $match: {
      "items.price": { $gte: 15000 },
      "items.product": { $in: ["Mobile", "Laptop"] }
    }
  },
  {
    $group: {
      _id: "$items.product",
      numberOfOrders: { $sum: 1 },
      totalRevenue: { $sum: "$items.price" },
      averagePrice: { $avg: "$items.price" },
      minimumPrice: { $min: "$items.price" },
      maximumPrice: { $max: "$items.price" },
      customerList: { $push: "$name" }
    }
  },
  {
```

```

    $project: {
      _id: 0,
      product: "$_id",
      numberOfOrders: 1,
      totalRevenue: 1,
      averagePrice: 1,
      minimumPrice: 1,
      maximumPrice: 1,
      customerList: 1
    }
  },
  {
    $out: "high_value_products_report"
  }
])
<
> db.high_value_products_report.find()
< {
  _id: ObjectId('6a11bdbec1680f7e726a327e'),
  numberOfOrders: 2,
  totalRevenue: 30000,
  averagePrice: 15000,
  minimumPrice: 15000,
  maximumPrice: 15000
}

```

```
> db.orders.aggregate([
  {
    $unwind: "$items"
  },
  {
    $group: {
      _id: "$name",
      totalProductsPurchased: { $sum: 1 },
      totalAmountSpent: { $sum: "$items.price" },
      averagePrice: { $avg: "$items.price" },
      minimumPrice: { $min: "$items.price" },
      maximumPrice: { $max: "$items.price" },
      listOfProducts: { $push: "$items.product" }
    }
  },
  {
    $match: {
      totalProductsPurchased: { $gt: 1 },
      totalAmountSpent: { $gte: 30000 }
    }
  },
  {
    $project: {
```

```
$project: {
  _id: 0,
  customerName: "$_id",
  totalAmountSpent: 1,
  averagePrice: 1,
  minimumPrice: 1,
  maximumPrice: 1,
  listOfProducts: 1
}
```

```
},
```

```
{
```

```
  $out: "customer_spending_report"
```

```
}
```

```
])
```

```
<
```

```
> db.customer_spending_report.find()
```

```
< {
```

```
  _id: ObjectId('6a11c276c1680f7e726a3280'),
```

```
  totalAmountSpent: 52000,
```

```
  averagePrice: 26000,
```

```
  minimumPrice: 2000,
```

```
  maximumPrice: 50000,
```

```
  listOfProducts: [
```

```
    'Earphones',
```

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CONNECTIONS (1)

Search connections

localhost:27017

admin

config

hospital

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online_shopping

tickets

bookings

customers

movies

>_MONGOSH

> use tickets

< switched to db tickets

> db.createCollection("customers")

< { ok: 1 }

> db.createCollection("movies")

< { ok: 1 }

> db.createCollection("bookings")

< { ok: 1 }

> db.customers.insertMany([

{

_id: 1,

name: "Arun",

profile: { age: 25, city: "Chennai" } // One-to-One

}

,

{

_id: 2,

name: "Priya",

profile: { age: 22, city: "Bangalore" }

}

,

{

_id: 3,

name: "Rahul",

profile: { age: 30, city: "Hyderabad" }

}

])


```
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5
  }
}

> db.movies.insertMany([
  { _id: "M1", title: "Inception", genre: "Sci-Fi", price: 300 },
  { _id: "M2", title: "Avatar", genre: "Fantasy", price: 350 },
  { _id: "M3", title: "Interstellar", genre: "Sci-Fi", price: 400 }
])

< {
  acknowledged: true,
  insertedIds: {
    '0': 'M1',
    '1': 'M2',
    '2': 'M3'
  }
}
```

```
> db.bookings.insertMany([
  { _id: 301, customer_id: 1, movie_id: "M1", show_date: "2026-02-01" },
  { _id: 302, customer_id: 1, movie_id: "M2", show_date: "2026-02-03" },
  { _id: 303, customer_id: 2, movie_id: "M1", show_date: "2026-02-05" },
  { _id: 304, customer_id: 3, movie_id: "M3", show_date: "2026-02-06" },
  { _id: 305, customer_id: 3, movie_id: "M1", show_date: "2026-02-08" },
  { _id: 306, customer_id: 4, movie_id: "M2", show_date: "2026-02-10" },
  { _id: 307, customer_id: 5, movie_id: "M3", show_date: "2026-02-12" },
  { _id: 308, customer_id: 5, movie_id: "M2", show_date: "2026-02-14" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 301,
    '1': 302,
    '2': 303,
    '3': 304,
    '4': 305,
    '5': 306,
    '6': 307,
    '7': 308
  }
}
```

```
> db.bookings.aggregate([
  {$group: {_id: "$movie",totalBookings: { $sum: 1 }}}},
  {$project: {_id: 0,movie: "$_id",totalBookings: 1}}
])
< {
  totalBookings: 8,
  movie: null
}
> db.bookings.aggregate([
  {$group: {_id: "$movie_id",totalBookings: { $sum: 1 }}}},
  {$project: {_id: 0,movie_id: "$_id",totalBookings: 1}}
])
< {
  totalBookings: 3,
  movie_id: 'M2'
}
{
  totalBookings: 3,
  movie_id: 'M1'
}
{
  totalBookings: 2,
  movie_id: 'M3'
```

```
> db.movies.aggregate([
  {$lookup: {from: "bookings",localField: "_id",foreignField: "movie_id",as: "bookingDetails"}},
  {$unwind: "$bookingDetails"},
  {$group: {_id: "$title",totalTicketsSold: { $sum: 1 },
    totalRevenue: { $sum: "$price" },
    averageTicketPrice: { $avg: "$price" },
    minimumTicketPrice: { $min: "$price" },
    maximumTicketPrice: { $max: "$price" }}}},
  {$project: {_id: 0,title: "$_id",totalTicketsSold: 1,
    totalRevenue: 1,
    averageTicketPrice: 1,
    minimumTicketPrice: 1,
    maximumTicketPrice: 1}}
])
< {
  totalTicketsSold: 3,
  totalRevenue: 1050,
  averageTicketPrice: 350,
  minimumTicketPrice: 350,
  maximumTicketPrice: 350,
  title: 'Avatar'
}
{
```

```

> db.customers.aggregate([
  {$lookup: {from: "bookings",localField: "_id",foreignField: "customer_id",as: "bookingDetails"}},
  {$unwind: "$bookingDetails"},
  {$lookup: {from: "movies",localField: "bookingDetails.movie_id",foreignField: "_id",as: "movieDetails"}},
  {$unwind: "$movieDetails"},
  {$group: {_id: "$name",
    numberOfMoviesBooked: { $sum: 1 },
    totalAmountSpent: { $sum: "$movieDetails.price" },
    averageSpending: { $avg: "$movieDetails.price" },
    listOfMoviesWatched: { $push: "$movieDetails.title" } }},
  {$project: {_id: 0,
    customerName: "$_id",
    numberOfMoviesBooked: 1,
    totalAmountSpent: 1,
    averageSpending: 1,
    listOfMoviesWatched: 1}}
])
< {
  numberOfMoviesBooked: 1,
  totalAmountSpent: 350,
  averageSpending: 350,
  listOfMoviesWatched: [
    'Avatar'
  ]
}

```



Search



ENG
IN



```
> db.bookings.aggregate([
  {$group: {_id: "$customer_id",totalBookings: { $sum: 1 }}}},
  {$match: {totalBookings: { $gt: 1 }}}
])
< {
  _id: 3,
  totalBookings: 2
}
{
  _id: 1,
  totalBookings: 2
}
{
  _id: 5,
  totalBookings: 2
}
```

```
> db.movies.aggregate([
  {$lookup: {from: "bookings",localField: "_id",foreignField: "movie_id",as: "bookingDetails"}},
  {$unwind: "$bookingDetails"},
  {$group: {_id: "$genre",totalBookings: { $sum: 1 }}}},
  {$project: {_id: 0, genre: "$_id",totalBookings: 1}}
])
< {
  totalBookings: 3,
  genre: 'Fantasy'
}
{
  totalBookings: 5,
  genre: 'Sci-Fi'
}
```



```

> db.movies.aggregate([
  {$lookup: {from: "bookings",localField: "_id",foreignField: "movie_id",as: "bookingDetails"}},
  {$unwind: "$bookingDetails"},
  {$lookup: {from: "customers",localField: "bookingDetails.customer_id",foreignField: "_id",as: "customerDetails"}},
  {$unwind: "$customerDetails"},
  {$group: {_id: "$title",totalNumberOfBookings: { $sum: 1 },
    listOfCustomers: { $push: "$customerDetails.name"}}},
  {$project: [_id: 0, movie: "$_id",
    totalNumberOfBookings: 1,
    listOfCustomers: 1]}
])
< {
  totalNumberOfBookings: 2,
  listOfCustomers: [
    'Rahul',
    'Kiran'
  ],
  movie: 'Interstellar'
}
{
  totalNumberOfBookings: 3,
  listOfCustomers: [
    'Arun',

```